NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-18546/2B
27 April 1966
SUPERSEDING
MIL-R-18546/2A
13 July 1965
(See 6.5)

PERFORMANCE SPECIFICATION

RESISTORS, FIXED, WIRE-WOUND

(POWER TYPE, CHASSIS MOUNTED),

STYLES RE77 AND RE80

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for styles RE77 and RE80 resistors.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-18546 - Resistors, Fixed, Wire-wound (Power Type, Chassis Mounted), General Specification for.

STANDARD

MILITARY

MS35690 - Nut, Plain, Hexagon, UNC-2B and UNF-2B.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-18546, and as specified herein.

3.2 Design and construction. The resistors shall be of the design, construction, and physical dimensions specified on figure 1, and as specified herein.

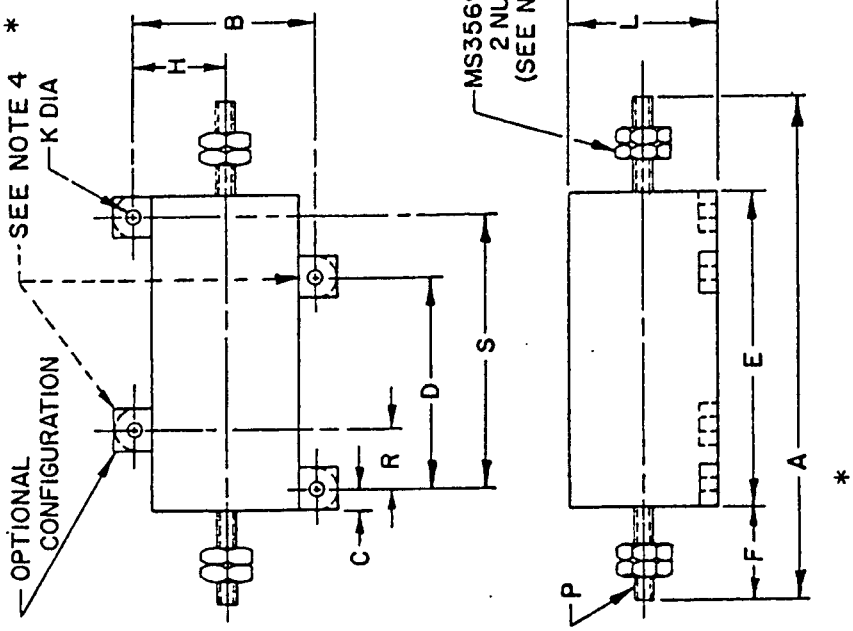
3.2.1 Weight. The maximum weight shall be as specified in table I.

TABLE I. Weight.

Resistor style	Weight Grams, maximum	
	Characteristic G	Characteristic N
RE77 - - - - - - - - - - - - -	400	440
RE80 - - - - - - - - - - - - -	800	880

FSC 5905

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:42Z
Check the source to verify that this is the current version before use.



Resistor style	A	B	C±.031 (.79)	D	E±.094 (2.39)	F	G±.031 (.79)	H±.031 (.79)	J±.031 (.79)	K±.010 (.25)	L±.031 (.79)	M±.062 (1.57)	N±.031 (.79)	P	R±.010 (.25)	S±.010 (.25)
RE77	5.478±.094 (139.14±2.39)	2.250±.010 (57.15±.25)	.375 (9.53)		3.500 (88.90)	.989±.031 (25.12±.79)	1.812 (46.02)	1.125 (28.58)	2.812 (71.42)	.188 (4.78)	1.750 (44.45)	.770 (19.56)	.188 (4.78)	12-24 UNC-2A		2.750 (69.85)
RE80	7.000±.125 (177.80±3.18)	2.500±.015 (63.50±.40)	.312 (7.93)	3.000±.010 (76.20±.25)	4.500 (114.30)	1.250±.062 (31.75±1.57)	2.125 (53.98)	1.250 (31.75)	3.000 (76.20)	.188 (4.78)	2.188 (55.58)	1.000 (25.40)	.250 (6.35)	1/4-20 UNC-2A	.875 (22.23)	3.875 (98.43)

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. MS35690-411 applies to RE80 only.
4. Mounting tabs apply to RE80 only.
5. Millimeters are in parentheses.

FIGURE 1. Styles RE77 and RE80 resistors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:42Z
Check the source to verify that this is the current version before use.

3.3 Power rating. The power rating shall be as specified in table II, based on full-load operation at an ambient temperature of 25° C, when mounted on the chassis specified in table V.

TABLE II. Power rating.

Resistor style	Power rating	Free air 1/
RE77 - - - - - - - - - - - - - - -	75	30
RE80 - - - - - - - - - - - - - - -	120	75

1/ Free air power ratings listed for information only.

3.4 Resistance. The minimum and maximum nominal resistance values shall be as specified in table III.

TABLE III. Resistance values.

Resistor style	Minimum		Maximum	
	Characteristic G	Characteristic N	Characteristic G 1/	Characteristic N
	ohms	ohms	ohms	ohms
RE77 - - - -	.05	1	29,400	14,700
RE80 - - - -	.10	1	35,700	17,400

1/ Based on use of 0.00175 inch nominal diameter wire.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-18546, and as specified herein.

4.2 Solderability. This test is not applicable to this specification.

4.3 Terminal strength. The terminal strength and applied torque shall be as specified in table IV.

TABLE IV. Terminal strength.

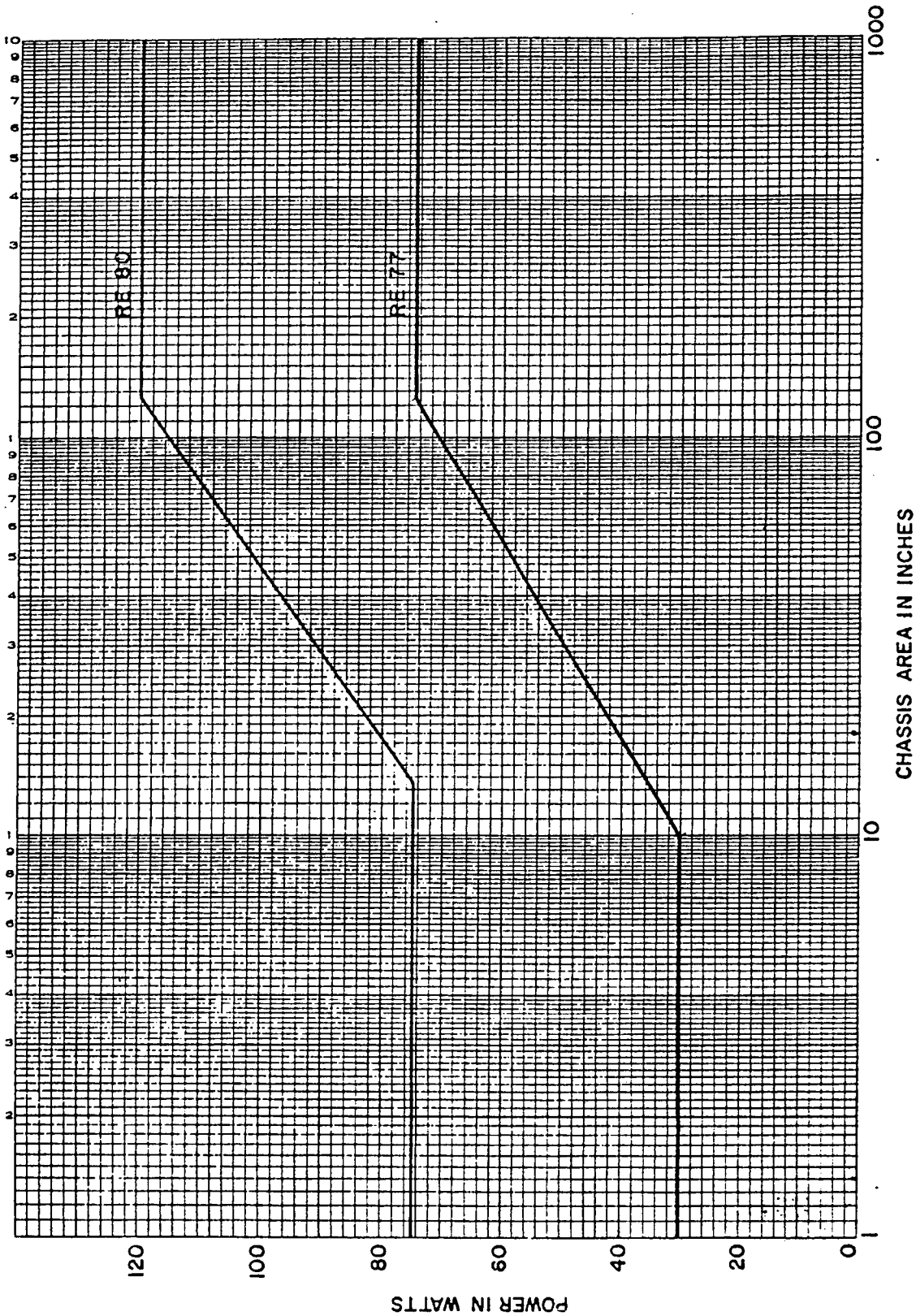
Resistor style	Direct pull	Torque
	Pounds	Pounds-Inches
RE77 - - - - - - - - - - - - - - -	$10\ ^{+0}_{-1/2}$	24
RE80 - - - - - - - - - - - - - - -	$10\ ^{+0}_{-1/2}$	32

4.4 Dielectric withstanding voltage.

4.4.1 At atmospheric pressure. The magnitude of the test voltage shall be 4,500 volts.

4.4.2 At reduced barometric pressure. The magnitude of the test voltage shall be 1,000 volts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:42Z
Check the source to verify that this is the current version before use.



NOTES: 1. The chassis derating curves are based on the full power ratings at an ambient temperature of 25°C.
2. These curves are independent of the temperature directing curves.

FIGURE 2. Chassis-area derating curves.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:42Z
Check the source to verify that this is the current version before use.

4.5 Life. The chassis dimensions shall be as specified in table V.

TABLE V. Chassis dimensions.

Resistor style	Length, width, and height	Thickness
	Inches	Inch
RE77 - - - - - - - - - - - - - - -	9 x 7 x 2	0.06
RE80 - - - - - - - - - - - - - - -	9 x 7 x 2	0.06

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-18546.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-18546 are applicable to this specification.

6.2 Power rating (free air). The free-air resistor (not mounted on a chassis) shall be as specified in table II.

6.3 Chassis-area derating curve. Figure 2 may be used for design information.

* 6.4 Interchangeability. Items in this specification are mutually interchangeable with items of the same style and characteristic procured under MIL-R-18546/2A.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - MU, MI, SM, EL
Navy - SH
Air Force - 11, 17, 85

User activities:
Navy - WP, MC
Air Force - 14, 19

Preparing activity:
Navy - SH

(Project 5905-0487-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:42Z
Check the source to verify that this is the current version before use.